Everyday Math

197. **Paycheck** Annie has two jobs. She gets paid $\$14.04$ per hour for tutoring at City College and $\$8.75$ per hour at a coffee shop. Last week she tutored for 8 hours and worked at the coffee shop for 15 hours.
ⓐ How much did she earn?
ⓑ If she had worked all 23 hours as a tutor instead of working both jobs, how much more would she have earned?

198. **Paycheck** Jake has two jobs. He gets paid $\$7.95$ per hour at the college cafeteria and $\$20.25$ at the art gallery. Last week he worked 12 hours at the cafeteria and 5 hours at the art gallery.
ⓐ How much did he earn?
ⓑ If he had worked all 17 hours at the art gallery instead of working both jobs, how much more would he have earned?

Writing Exercises

199. At the 2010 winter Olympics, two skiers took the silver and bronze medals in the Men's Super-G ski event. Miller's time was 1 minute 30.62 seconds and Weibrecht's time was 1 minute 30.65 seconds. Find the difference in their times and then write the name of that decimal.

200. Find the quotient of $0.12 \div 0.04$ and explain in words all the steps taken.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
add and subtract decimals.			
multiply decimals.			
divide decimals.			
use decimals in money applications.			

ⓑ *After reviewing this checklist, what will you do to become confident for all objectives?*

5.3 Decimals and Fractions

Learning Objectives

By the end of this section, you will be able to:

- Convert fractions to decimals
- Order decimals and fractions
- Simplify expressions using the order of operations
- Find the circumference and area of circles

☑ **BE PREPARED** 5.7 Before you get started, take this readiness quiz.

Divide: $0.24 \div 8$.
If you missed this problem, review Example 5.19.

☑ **BE PREPARED** 5.8 Order $0.64 __ 0.6$ using $<$ or $>$.
If you missed this problem, review Example 5.7.

☑ **BE PREPARED** 5.9 Order $-0.2 __ -0.1$ using $<$ or $>$.
If you missed this problem, review Example 5.8.

Convert Fractions to Decimals

In Decimals, we learned to convert decimals to fractions. Now we will do the reverse—convert fractions to decimals. Remember that the fraction bar indicates division. So $\frac{4}{5}$ can be written $4 \div 5$ or $5\overline{)4}$. This means that we can convert a

fraction to a decimal by treating it as a division problem.

Convert a Fraction to a Decimal

To convert a fraction to a decimal, divide the numerator of the fraction by the denominator of the fraction.

EXAMPLE 5.28

Write the fraction $\frac{3}{4}$ as a decimal.

Solution

A fraction bar means division, so we can write the fraction $\frac{3}{4}$ using division.	$4\overline{)3}$
Divide.	$\begin{array}{r} 0.75 \\ 4\overline{)3.00} \\ \underline{28} \\ 20 \\ \underline{20} \\ 0 \end{array}$
	So the fraction $\frac{3}{4}$ is equal to 0.75.

TRY IT 5.55 Write the fraction as a decimal: $\frac{1}{4}$.

TRY IT 5.56 Write the fraction as a decimal: $\frac{3}{8}$.

EXAMPLE 5.29

Write the fraction $-\frac{7}{2}$ as a decimal.

Solution

The value of this fraction is negative. After dividing, the value of the decimal will be negative. We do the division ignoring the sign, and then write the negative sign in the answer.	$-\frac{7}{2}$
Divide 7 by 2.	$\begin{array}{r} 3.5 \\ 2\overline{)7.0} \\ \underline{6} \\ 10 \\ \underline{10} \\ 0 \end{array}$
	So, $-\frac{7}{2} = -3.5$.

TRY IT 5.57 Write the fraction as a decimal: $-\frac{9}{4}$.

TRY IT 5.58 Write the fraction as a decimal: $-\frac{11}{2}$.

Repeating Decimals

So far, in all the examples converting fractions to decimals the division resulted in a remainder of zero. This is not always the case. Let's see what happens when we convert the fraction $\frac{4}{3}$ to a decimal. First, notice that $\frac{4}{3}$ is an improper fraction. Its value is greater than 1. The equivalent decimal will also be greater than 1.

We divide 4 by 3.

$$
\begin{array}{r}
1.333\ldots \\
3\overline{)4.000} \\
\underline{3} \\
10 \\
\underline{9} \\
10 \\
\underline{9} \\
10 \\
\underline{9} \\
1
\end{array}
$$

No matter how many more zeros we write, there will always be a remainder of 1, and the threes in the quotient will go on forever. The number $1.333\ldots$ is called a repeating decimal. Remember that the "..." means that the pattern repeats.

Repeating Decimal

A **repeating decimal** is a decimal in which the last digit or group of digits repeats endlessly.

How do you know how many 'repeats' to write? Instead of writing $1.333\ldots$ we use a shorthand notation by placing a line over the digits that repeat. The repeating decimal $1.333\ldots$ is written $1.\overline{3}$. The line above the 3 tells you that the 3 repeats endlessly. So $1.333\ldots = 1.\overline{3}$

For other decimals, two or more digits might repeat. Table 5.5 shows some more examples of repeating decimals.

$1.333\ldots = 1.\overline{3}$	3 is the repeating digit
$4.1666\ldots = 4.1\overline{6}$	6 is the repeating digit
$4.161616\ldots = 4.\overline{16}$	16 is the repeating block
$0.271271271\ldots = 0.\overline{271}$	271 is the repeating block

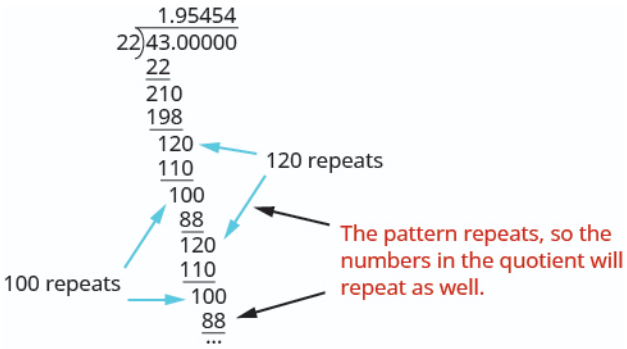
Table 5.5

EXAMPLE 5.30

Write $\frac{43}{22}$ as a decimal.

Solution

Divide 43 by 22.



Notice that the differences of 120 and 100 repeat, so there is a repeat in the digits of the quotient; 54 will repeat endlessly. The first decimal place in the quotient, 9, is not part of the pattern. So,

$$\frac{43}{22} = 1.9\overline{54}$$

TRY IT 5.59 Write as a decimal: $\frac{27}{11}$.

TRY IT 5.60 Write as a decimal: $\frac{51}{22}$.

It is useful to convert between fractions and decimals when we need to add or subtract numbers in different forms. To add a fraction and a decimal, for example, we would need to either convert the fraction to a decimal or the decimal to a fraction.

EXAMPLE 5.31

Simplify: $\frac{7}{8} + 6.4$.

Solution

		$\frac{7}{8} + 6.4$
Change $\frac{7}{8}$ to a decimal.	$8\overline{)7.000}$ 64 60 56 40 40 0	$0.875 + 6.4$
Add.		7.275

TRY IT 5.61 Simplify: $\frac{3}{8} + 4.9$.

TRY IT 5.62 Simplify: $5.7 + \frac{13}{20}$.

Order Decimals and Fractions

In Decimals, we compared two decimals and determined which was larger. To compare a decimal to a fraction, we will first convert the fraction to a decimal and then compare the decimals.

EXAMPLE 5.32

Order $\frac{3}{8}$ __ 0.4 using $<$ or $>$.

Solution

	$\frac{3}{8}$ __ 0.4
Convert $\frac{3}{8}$ to a decimal.	0.375 __ 0.4

Compare 0.375 to 0.4	$0.375 < 0.4$
Rewrite with the original fraction.	$\frac{3}{8} < 0.4$

TRY IT 5.63 Order each of the following pairs of numbers, using $<$ or $>$.

$\frac{17}{20}$ __ 0.82

TRY IT 5.64 Order each of the following pairs of numbers, using $<$ or $>$.

$\frac{3}{4}$ __ 0.785

When ordering negative numbers, remember that larger numbers are to the right on the number line and any positive number is greater than any negative number.

EXAMPLE 5.33

Order -0.5 ___ $-\frac{3}{4}$ using $<$ or $>$.

Solution

	-0.5 ___ $-\frac{3}{4}$
Convert $-\frac{3}{4}$ to a decimal.	-0.5 ___ -0.75
Compare -0.5 to -0.75.	$-0.5 > -0.75$
Rewrite the inequality with the original fraction.	$-0.5 > -\frac{3}{4}$

TRY IT 5.65 Order each of the following pairs of numbers, using $<$ or $>$:

$-\frac{5}{8}$ __ -0.58

TRY IT 5.66 Order each of the following pairs of numbers, using $<$ or $>$:

-0.53 __ $-\frac{11}{20}$

EXAMPLE 5.34

Write the numbers $\frac{13}{20}, 0.61, \frac{11}{16}$ in order from smallest to largest.

Solution

	$\frac{13}{20}, 0.61, \frac{11}{16}$
Convert the fractions to decimals.	$0.65, 0.61, 0.6875$
Write the smallest decimal number first.	$0.61,$ ____, ____

Write the next larger decimal number in the middle place.	$0.61, 0.65, _____$
Write the last decimal number (the larger) in the third place.	$0.61, 0.65, 0.6875$
Rewrite the list with the original fractions.	$0.61, \frac{13}{20}, \frac{11}{16}$

TRY IT 5.67 Write each set of numbers in order from smallest to largest: $\frac{7}{8}, \frac{4}{5}, 0.82$.

TRY IT 5.68 Write each set of numbers in order from smallest to largest: $0.835, \frac{13}{16}, \frac{3}{4}$.

Simplify Expressions Using the Order of Operations

The order of operations introduced in Use the Language of Algebra also applies to decimals. Do you remember what the phrase "Please excuse my dear Aunt Sally" stands for?

EXAMPLE 5.35

Simplify the expressions:

ⓐ $7(18.3 - 21.7)$ ⓑ $\frac{2}{3}(8.3 - 3.8)$

Solution

ⓐ

	$7(18.3 - 21.7)$
Simplify inside parentheses.	$7(-3.4)$
Multiply.	-23.8

ⓑ

	$\frac{2}{3}(8.3 - 3.8)$
Simplify inside parentheses.	$\frac{2}{3}(4.5)$
Write 4.5 as a fraction.	$\frac{2}{3}(\frac{4.5}{1})$
Multiply.	$\frac{9}{3}$
Simplify.	3

TRY IT 5.69 Simplify: ⓐ $8(14.6 - 37.5)$ ⓑ $\frac{3}{5}(9.6 - 2.1)$.

TRY IT 5.70 Simplify: ⓐ $25(25.69 - 56.74)$ ⓑ $\frac{2}{7}(11.9 - 4.2)$.

EXAMPLE 5.36

Simplify each expression:

ⓐ $6 \div 0.6 + (0.2)4 - (0.1)^2$ ⓑ $\left(\frac{1}{10}\right)^2 + (3.5)(0.9)$

Solution

ⓐ

	$6 \div 0.6 + (0.2)4 - (0.1)^2$
Simplify exponents.	$6 \div 0.6 + (0.2)4 - 0.01$
Divide.	$10 + (0.2)4 - 0.01$
Multiply.	$10 + 0.8 - 0.01$
Add.	$10.8 - 0.01$
Subtract.	10.79

ⓑ

	$\left(\frac{1}{10}\right)^2 + (3.5)(0.9)$
Simplify exponents.	$\frac{1}{100} + (3.5)(0.9)$
Multiply.	$\frac{1}{100} + 3.15$
Convert $\frac{1}{100}$ to a decimal.	$0.01 + 3.15$
Add.	3.16

TRY IT 5.71 Simplify: $9 \div 0.9 + (0.4)3 - (0.2)^2$.

TRY IT 5.72 Simplify: $\left(\frac{1}{2}\right)^2 + (0.3)(4.2)$.

Find the Circumference and Area of Circles

The properties of circles have been studied for over 2,000 years. All circles have exactly the same shape, but their sizes are affected by the length of the **radius**, a line segment from the center to any point on the circle. A line segment that passes through a circle's center connecting two points on the circle is called a **diameter**. The diameter is twice as long as the radius. See Figure 5.6.

The size of a circle can be measured in two ways. The distance around a circle is called its **circumference**.

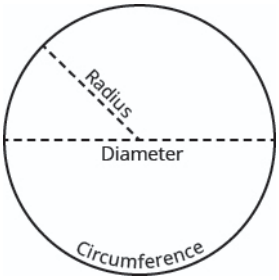


Figure 5.6

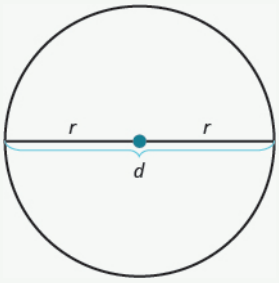
Archimedes discovered that for circles of all different sizes, dividing the circumference by the diameter always gives the same number. The value of this number is pi, symbolized by Greek letter π (pronounced pie). However, the exact value of π cannot be calculated since the decimal never ends or repeats (we will learn more about numbers like this in The Properties of Real Numbers.)

MANIPULATIVE MATHEMATICS

Doing the Manipulative Mathematics activity Pi Lab will help you develop a better understanding of pi.

If we want the exact circumference or area of a circle, we leave the symbol π in the answer. We can get an approximate answer by substituting 3.14 as the value of π. We use the symbol $\approx$ to show that the result is approximate, not exact.

Properties of Circles



r is the length of the radius.
d is the length of the diameter.

The circumference is $2\pi r$. $C = 2\pi r$

The area is πr^2. $A = \pi r^2$

Since the diameter is twice the radius, another way to find the circumference is to use the formula $C = \pi d$.

Suppose we want to find the exact area of a circle of radius 10 inches. To calculate the area, we would evaluate the formula for the area when $r = 10$ inches and leave the answer in terms of π.

$$A = \pi r^2$$
$$A = \pi(10^2)$$
$$A = \pi \cdot 100$$

We write π after the 100. So the exact value of the area is $A = 100\pi$ square inches.

To approximate the area, we would substitute $\pi \approx 3.14$.

$$A = 100\pi$$
$$\approx 100 \cdot 3.14$$
$$\approx 314 \text{ square inches}$$

Remember to use square units, such as square inches, when you calculate the area.

EXAMPLE 5.37

A circle has radius 10 centimeters. Approximate its (a) circumference and (b) area.

Solution

(a) Find the circumference when $r = 10$.

Write the formula for circumference.	$C = 2\pi r$
Substitute 3.14 for π and 10 for ,r.	$C \approx 2(3.14)(10)$
Multiply.	$C \approx 62.8$ centimeters

(b) Find the area when $r = 10$.

Write the formula for area.	$A = \pi r^2$
Substitute 3.14 for π and 10 for r.	$A \approx (3.14)(10)^2$
Multiply.	$A \approx 314$ square centimeters

TRY IT 5.73 A circle has radius 50 inches. Approximate its (a) circumference and (b) area.

TRY IT 5.74 A circle has radius 100 feet. Approximate its (a) circumference and (b) area.

EXAMPLE 5.38

A circle has radius 42.5 centimeters. Approximate its (a) circumference and (b) area.

Solution

(a) Find the circumference when $r = 42.5$.

Write the formula for circumference.	$C = 2\pi r$
Substitute 3.14 for π and 42.5 for r	$C \approx 2(3.14)(42.5)$
Multiply.	$C \approx 266.9$ centimeters

(b) Find the area when $r = 42.5$.

Write the formula for area.	$A = \pi r^2$

Substitute 3.14 for π and 42.5 for r.	$A \approx (3.14)(42.5)^2$
Multiply.	$A \approx 5671.625$ square centimeters

TRY IT 5.75 A circle has radius 51.8 centimeters. Approximate its (a) circumference and (b) area.

TRY IT 5.76 A circle has radius 26.4 meters. Approximate its (a) circumference and (b) area.

Approximate π with a Fraction

Convert the fraction $\frac{22}{7}$ to a decimal. If you use your calculator, the decimal number will fill up the display and show 3.14285714. But if we round that number to two decimal places, we get 3.14, the decimal approximation of π. When we have a circle with radius given as a fraction, we can substitute $\frac{22}{7}$ for π instead of 3.14. And, since $\frac{22}{7}$ is also an approximation of π, we will use the ≈ symbol to show we have an approximate value.

EXAMPLE 5.39

A circle has radius $\frac{14}{15}$ meter. Approximate its (a) circumference and (b) area.

Solution

(a) Find the circumference when $r = \frac{14}{15}$.

Write the formula for circumference.	$C = 2\pi r$
Substitute $\frac{22}{7}$ for π and $\frac{14}{15}$ for r.	$C \approx 2(\frac{22}{7})(\frac{14}{15})$
Multiply.	$C \approx \frac{88}{15}$ meters

(b) Find the area when $r = \frac{14}{15}$.

Write the formula for area.	$A = \pi r^2$
Substitute $\frac{22}{7}$ for π and $\frac{14}{15}$ for r.	$A \approx (\frac{22}{7})(\frac{14}{15})^2$
Multiply.	$A \approx \frac{616}{225}$ square meters

TRY IT 5.77 A circle has radius $\frac{5}{21}$ meters. Approximate its (a) circumference and (b) area.

TRY IT 5.78 A circle has radius $\frac{10}{33}$ inches. Approximate its (a) circumference and (b) area.

MEDIA

ACCESS ADDITIONAL ONLINE RESOURCES

Converting a Fraction to a Decimal - Part 2 (https://www.openstax.org/l/24convfrac2dec)
Convert a Fraction to a Decimal (repeating) (https://www.openstax.org/l/24convfr2decrep)
Compare Fractions and Decimals using Inequality Symbols (https://www.openstax.org/l/24compfrcdec)

Determine the Area of a Circle (https://www.openstax.org/l/24circlearea)
Determine the Circumference of a Circle (https://www.openstax.org/l/24circlecirc)

SECTION 5.3 EXERCISES

Practice Makes Perfect

Convert Fractions to Decimals

In the following exercises, convert each fraction to a decimal.

201. $\frac{2}{5}$

202. $\frac{4}{5}$

203. $-\frac{3}{8}$

204. $-\frac{5}{8}$

205. $\frac{17}{20}$

206. $\frac{13}{20}$

207. $\frac{11}{4}$

208. $\frac{17}{4}$

209. $-\frac{310}{25}$

210. $-\frac{284}{25}$

211. $\frac{5}{9}$

212. $\frac{2}{9}$

213. $\frac{15}{11}$

214. $\frac{18}{11}$

215. $\frac{15}{111}$

216. $\frac{25}{111}$

In the following exercises, simplify the expression.

217. $\frac{1}{2} + 6.5$

218. $\frac{1}{4} + 10.75$

219. $2.4 + \frac{5}{8}$

220. $3.9 + \frac{9}{20}$

221. $9.73 + \frac{17}{20}$

222. $6.29 + \frac{21}{40}$

Order Decimals and Fractions

In the following exercises, order each pair of numbers, using < or >.

223. $\frac{1}{8}$ ___ 0.8

224. $\frac{1}{4}$ ___ 0.4

225. $\frac{2}{5}$ ___ 0.25

226. $\frac{3}{5}$ ___ 0.35

227. 0.725 ___ $\frac{3}{4}$

228. 0.92 ___ $\frac{7}{8}$

229. 0.66 ___ $\frac{2}{3}$

230. 0.83 ___ $\frac{5}{6}$

231. -0.75 ___ $-\frac{4}{5}$

232. -0.44 ___ $-\frac{9}{20}$

233. $-\frac{3}{4}$ ___ -0.925

234. $-\frac{2}{3}$ ___ -0.632

In the following exercises, write each set of numbers in order from least to greatest.

235. $\frac{3}{5}, \frac{9}{16}, 0.55$

236. $\frac{3}{8}, \frac{7}{20}, 0.36$

237. $0.702, \frac{13}{20}, \frac{5}{8}$

238. $0.15, \frac{3}{16}, \frac{1}{5}$

239. $-0.3, -\frac{1}{3}, -\frac{7}{20}$

240. $-0.2, -\frac{3}{20}, -\frac{1}{6}$

241. $-\frac{3}{4}, -\frac{7}{9}, -0.7$

242. $-\frac{8}{9}, -\frac{4}{5}, -0.9$

Simplify Expressions Using the Order of Operations

In the following exercises, simplify.

243. $10(25.1 - 43.8)$

244. $30(18.1 - 32.5)$

245. $62(9.75 - 4.99)$

246. $42(8.45 - 5.97)$

247. $\frac{3}{4}(12.4 - 4.2)$

248. $\frac{4}{5}(8.6 + 3.9)$

249. $\frac{5}{12}(30.58 + 17.9)$

250. $\frac{9}{16}(21.96 - 9.8)$

251. $10 \div 0.1 + (1.8)4 - (0.3)^2$

252. $5 \div 0.5 + (3.9)6 - (0.7)^2$

253. $(37.1 + 52.7) \div (12.5 \div 62.5)$

254. $(11.4 + 16.2) \div (18 \div 60)$

255. $\left(\frac{1}{5}\right)^2 + (1.4)(6.5)$

256. $\left(\frac{1}{2}\right)^2 + (2.1)(8.3)$

257. $-\frac{9}{10} \cdot \frac{8}{15} + 0.25$

258. $-\frac{3}{8} \cdot \frac{14}{15} + 0.72$

Mixed Practice

In the following exercises, simplify. Give the answer as a decimal.

259. $3\frac{1}{4} - 6.5$

260. $5\frac{2}{5} - 8.75$

261. $10.86 \div \frac{2}{3}$

262. $5.79 \div \frac{3}{4}$

263. $\frac{7}{8}(103.48) + 1\frac{1}{2}(361)$

264. $\frac{5}{16}(117.6) + 2\frac{1}{3}(699)$

265. $3.6\left(\frac{9}{8} - 2.72\right)$

266. $5.1\left(\frac{12}{5} - 3.91\right)$

Find the Circumference and Area of Circles

In the following exercises, approximate the ⓐ circumference and ⓑ area of each circle. If measurements are given in fractions, leave answers in fraction form.

267. radius $= 5$ in.

268. radius $= 20$ in.

269. radius $= 9$ ft.

270. radius $= 4$ ft.

271. radius $= 46$ cm

272. radius $= 38$ cm

273. radius $= 18.6$ m

274. radius $= 57.3$ m

275. radius $= \frac{7}{10}$ mile

276. radius $= \frac{7}{11}$ mile

277. radius $= \frac{3}{8}$ yard

278. radius $= \frac{5}{12}$ yard

279. diameter $= \frac{5}{6}$ m

280. diameter $= \frac{3}{4}$ m

Everyday Math

281. Kelly wants to buy a pair of boots that are on sale for $\frac{2}{3}$ of the original price. The original price of the boots is $84.99. What is the sale price of the shoes?

282. An architect is planning to put a circular mosaic in the entry of a new building. The mosaic will be in the shape of a circle with radius of 6 feet. How many square feet of tile will be needed for the mosaic? (Round your answer up to the next whole number.)

Writing Exercises

283. Is it easier for you to convert a decimal to a fraction or a fraction to a decimal? Explain.

284. Describe a situation in your life in which you might need to find the area or circumference of a circle.

Self Check

ⓐ *After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.*

I can...	Confidently	With some help	No-I don't get it!
convert fractions to decimals.			
order decimals and fractions.			
simplify expressions using the order of operations.			
find the circumference and area of circles.			